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Energy Security Data Warehouse

*A Fact Constellation Architecture for Analyzing Geopolitical Risk and Energy Resilience
in Europe*

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Abstract

The energy crisis that followed Russia's invasion of Ukraine in 2022 showed how exposed Europe can be to geopolitical tensions. Gas and electricity prices rose sharply, but the impact was different from one country to another. Understanding these differences requires combining information on geopolitical risk, consumer energy prices, import dependency, and strategic reserves. In practice, these data come from different sources and are published at different time intervals, which makes direct comparison difficult.

This report describes the design and implementation of the **Energy Security Data Warehouse**. The system integrates seven datasets from the Federal Reserve Board and Eurostat covering the period from 1985 to 2026. It uses a Fact Constellation with four fact tables, linked by shared geographic and temporal dimensions, so that monthly, half-yearly, and annual observations retain their original level of detail.

The project also applies clear aggregation rules: prices, dependency rates, and geopolitical indices are not summed, while oil-stock levels are treated as semi-additive measures. A temporal bridge records changes in EU membership over time. The warehouse is loaded through an automated, idempotent Python ETL pipeline and checked with 12 data-quality tests. The analyses show a strong temporal association ($r > 0.80$) between geopolitical-risk indicators and retail tariffs in several European countries, and they support the assessment of oil-stock autonomy against the 90-day IEA benchmark.

Repository Information

Official Code Repository: <https://github.com/francescogallo2708-tech/energy-security-data-warehouse>

Environment: PostgreSQL 14+, Python 3.11+, psycpg2-binary, pandas, bash/PowerShell runners.

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1 Introduction, Scenario, and Research Objectives

1.1 Macroeconomic Context: The 2022 European Energy Crisis

Energy security has re-emerged as a paramount strategic concern for the European continent. For decades, the European Union’s industrial competitiveness and residential heating infrastructure relied heavily on stable, low-cost pipeline supplies of Russian natural gas. On February 24, 2022, the military invasion of Ukraine permanently fractured this supply paradigm. The subsequent weaponization of energy exports, pipeline curtailments, physical sabotage incidents (such as the Nord Stream explosions), and EU retaliatory sanction packages precipitated a supply contraction of historic proportions.

Between the first semester of 2021 and the second semester of 2022, natural gas wholesale prices at the Title Transfer Facility (TTF) benchmark in the Netherlands experienced an unprecedented ten-fold surge, rising from approximately €20/MWh to over €300/MWh. Under the European Union’s target electricity market design, where wholesale clearing prices are determined by the short-run marginal cost of the last generating unit dispatched (the merit-order effect), gas-fired power plants frequently set the clearing price. Consequently, wholesale gas inflation propagated instantaneously into retail consumer electricity bills.

However, the domestic ramifications of this macroeconomic shock diverged sharply across member states. Asymmetric geographical exposure, heterogeneous national generation portfolios (e.g., nuclear in France, hydro in the Nordics, coal in Poland), differing access to liquefied natural gas (LNG) regasification infrastructure, and divergent national fiscal subsidy packages created extreme divergence in consumer economic resilience.

1.2 The Analytical Challenge: Dimensional Fragmentation

Conducting rigorous, longitudinal evaluations of European energy resilience poses severe data engineering challenges. Analysts and policy researchers are routinely confronted with disparate information streams that exhibit structural incompatibilities:

1. **Incompatible Temporal Granularities:** Academic indices tracking international geopolitical threats are published at monthly resolution. Conversely, official Eurostat consumer utility price surveys are compiled on a semester basis (twice a year), national energy import dependency balances are compiled annually, and physical emergency oil reserves are monitored monthly.
2. **Complex Dimensional Classifications:** Tariff data is stratified across heterogeneous consumer bands (residential vs. industrial), tax exclusion tiers, and local national currencies.
3. **Shifting Political Geographies:** The legal boundaries of the European Union have evolved dynamically over the past four decades, incorporating new accessions (e.g., Eastern enlargement in 2004, Croatia in 2013) and witnessing the departure of the United Kingdom (Brexit on January 31, 2020).

1.3 Core Research Question and System Goals

To bridge these domains into a scientifically sound decision-support platform, this project addresses the following research question:

“To what extent, and through what temporal dynamics, do international geopolitical tensions co-vary with consumer energy prices, national import dependencies, and strategic oil stockpiles across European nations?”

The engineering objectives of the **Energy Security Data Warehouse** are:

- **Preserve Native Grain:** Refuse destructive synthetic interpolation or pre-aggregation of raw feeds, preserving the native atomic resolution of every business process.
- **Enforce Measure Algebra:** Strictly classify measures into additive, semi-additive, and non-additive categories, preventing false summations across dimensions.
- **Enable Seamless Drill-Across:** Provide conformed dimensions enabling cross-fact analytics across disparate business domains.
- **Support Reliable Loading:** Implement an idempotent, transaction-safe ETL pipeline with routine SQL checks after loading.

2 Theoretical Foundations of Data Warehousing

2.1 The Canonical Inmon Definition

As formalized by W. H. Inmon, a **Data Warehouse (DWH)** is:

“A subject-oriented, integrated, time-variant, and non-volatile collection of data in support of management’s decision-making process.”

Our architecture faithfully implements all four canonical pillars:

1. **Subject-Oriented:** The storage model is organized exclusively around analytical subjects of strategic importance (*Geopolitical Risk, Consumer Tariffs, Import Dependency, and Emergency Stocks*), rather than operational transactions.
2. **Integrated:** Structural and encoding heterogeneities across disparate sources are harmonized: country codes follow ISO-3166 standards, units of measure are mapped to physical standards, and currency conversions are decoupled.
3. **Time-Variant:** Every record is explicitly anchored to a temporal dimension spanning 1985 to 2026, preserving longitudinal historical truth rather than a current transactional snapshot.
4. **Non-Volatile:** Once validated and loaded via ACID transactions, data is immutable and accessed exclusively via read-only OLAP operators.

2.2 Systematic Comparison: OLTP vs. OLAP

The design addresses the fundamental requirements that differentiate decision-support systems from operational transactional databases ([Table 1](#)).

Dimension	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing)
Primary Goal	Real-time transactional execution	Strategic decision support and multi-decade intelligence
Schema Modeling	Highly normalized (3NF / BCNF)	Dimensional modeling (Star / Fact Constellation)
Query Patterns	High-frequency, discrete reads/writes (INSERT, UPDATE)	Complex, heavy read-only scans, aggregations, windowing
Granularity	Maximum atomic operational detail	Multi-level (from atomic facts to roll-up summaries)
Data Volatility	Dynamic, constant in-place state mutation	Non-volatile, append-only historical loading
Indexing	Targeted single-row primary key lookups	B-tree clustered indexes on foreign and surrogate keys

Table 1: Systematic comparison between operational OLTP and analytical OLAP paradigms.

2.3 Architectural Paradigms: ROLAP Implementation

Course theory classifies multidimensional implementations into three primary paradigms:

- **MOLAP (Multidimensional OLAP):** Uses proprietary hypercubes and sparse arrays. While delivering low query latency for static pre-computed views, MOLAP suffers from rigid recalculation costs, storage explosion under high dimensional sparsity, and commercial vendor lock-in.
- **ROLAP (Relational OLAP):** Maps multidimensional cubes directly to standard relational databases where fact cells correspond to tuples linked to dimension tables via foreign keys. It provides unlimited scalability, standard SQL compliance, ACID reliability, and dynamic indexing.
- **HOLAP (Hybrid OLAP):** Combines MOLAP pre-aggregates for top-level summaries with ROLAP relational tables for granular drill-downs.

This project implements a pure, production-grade **ROLAP architecture on PostgreSQL 14+**. By leveraging PostgreSQL’s robust query planner, advanced SQL:1999 window functions, and declarative partitioning capabilities, ROLAP provides transparent, fully auditable, and mathematically sound analytics without external proprietary engines.

3 Raw Data Sources and Data Profiling

The warehouse is populated exclusively from authoritative primary public repositories ([Table 2](#)). No synthetic, mock, or generated records are utilized.

Dataset	Agency / Authority	Native Grain	Horizon	Format	Raw Tuples
GPR Index	Fed Board (Caldara & Iacoviello)	Monthly	1985–2026	XLS	499 months
nrg_ind_id	Eurostat (Energy Dependency)	Annual	1990–2024	TSV	17,316
nrg_pc_202	Eurostat (Gas: Households)	Half-Yearly	2007–2025	TSV	60,526
nrg_pc_203	Eurostat (Gas: Industry)	Half-Yearly	2007–2025	TSV	109,038
nrg_pc_204	Eurostat (Electricity: Households)	Half-Yearly	2007–2025	TSV	65,466
nrg_pc_205	Eurostat (Electricity: Industry)	Half-Yearly	2007–2025	TSV	79,641
nrg_stk_oem	Eurostat (Emergency Oil Stocks)	Monthly	2013–2026	TSV	28,247

Table 2: Inventory of primary raw data feeds ingested into the data warehouse.

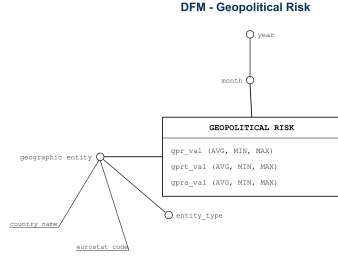
3.1 Data Profiling and Cleansing Challenges

Systematic profiling of the raw data feeds uncovered three critical engineering hurdles:

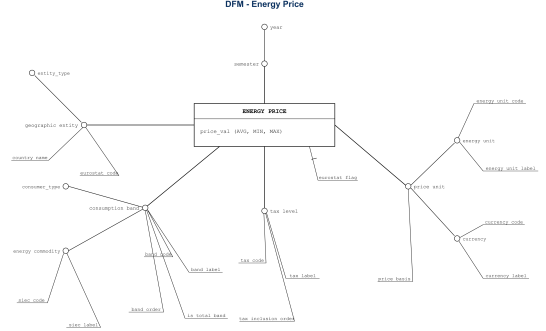
1. **Composite Matrix Headers (Wide TSV Format):** Eurostat TSV distributions package multi-attribute coordinates into a single comma-delimited string within the first column header (e.g., "`freq,siec,unit,geo\TIME_PERIOD`"). An automated vectorized Python unpivoting routine was engineered to unpivot these wide tables into normalized atomic tuples.
2. **Mixed Quality Flags and Special Characters:** Eurostat embeds qualitative status flags directly into numerical cells (e.g., "`0.1425 p`" for provisional, "`e`" for estimated, "`b`" for break in time series, "`:`" for unobserved). A regex tokenization engine isolates the clean floating-point measure while extracting and persisting the quality flag as a descriptive fact attribute.
3. **Validation of Negative Price Anomalies:** During profiling, negative consumer electricity prices were detected in the Netherlands and Czechia in 2022. Rather than discarding these as transmission artifacts, institutional verification confirmed that they represented emergency national tax credit rebates designed to buffer the gas crisis, which momentarily exceeded fixed network charges. The physical schema constraints were specifically configured to accommodate verified negative balances.

4 Conceptual Modeling: Dimensional Fact Model (DFM)

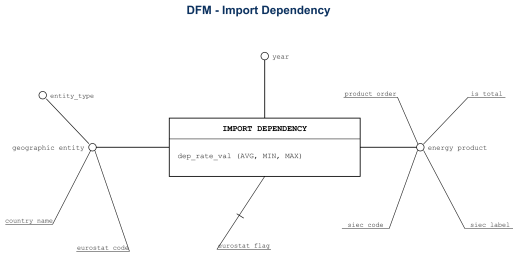
Conceptual modeling follows the **Dimensional Fact Model (DFM)** formalisms developed by Golfarelli, Maio, and Rizzi. The enterprise data warehouse is organized into four distinct fact schemes.



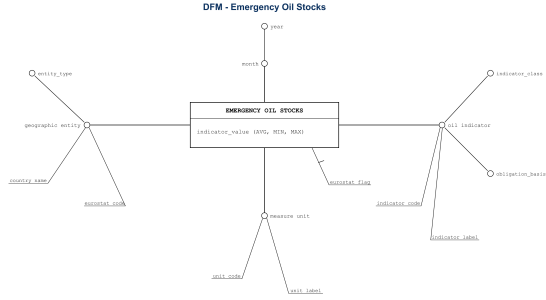
(a) FACT_GPR (Monthly)



(b) FACT_ENERGY_PRICE (Half-Yearly)



(c) FACT_IMPORT_DEPENDENCY (Annual)



(d) FACT_OIL_STOCKS (Monthly)

Figure 1: The four conceptual DFM schemes capturing native granularities, hierarchies, and measures.

4.1 Advanced DFM Constructs

The conceptual schemes incorporate advanced dimensional patterns:

- **Convergent DAG Time Hierarchies:** As illustrated in the temporal dimensions, the monthly ladder (Month $\rightarrow$ Quarter $\rightarrow$ Year) and the semester ladder (Semester $\rightarrow$ Year) converge into the common parent attribute Year. This creates a Directed Acyclic Graph (DAG) ensuring that separate facts can aggregate safely to a shared calendar milestone without violating dimensional roll-up consistency.
- **Degenerate Attributes:** Observational status flags (`eurostat_flag`) lack independent dimensional hierarchies and are modeled as degenerate attributes directly within the fact boxes.
- **Cross-Dimension Dependencies:** In the energy tariff fact scheme, consumption band definitions are strictly dependent on the commodity fuel category (gas bands D1–D3 vs. electricity bands I1–I6).

4.2 Rigorous Algebra of Measures

A cornerstone of this work is the strict classification of measures according to their mathematical properties of additivity (Table 3).

Measure Class	Mathematical Definition	Valid Operators	System Implementations
Non-Additive (<i>Unit / Ratio</i>)	Measures intensity or ratios. Non-summable across any dimension. $\sum$ is strictly prohibited.	AVG, MIN, MAX, or evaluating $\sum \frac{\text{num}}{\text{den}}$	Retail Price (€/kWh), Import Dependency (%), GPR Index
Semi-Additive (<i>Stock / Level</i>)	Measures physical balance at a specific time instant. Additive across space, non-additive across time .	Space: $\sum$ Time: AVG, MIN, MAX, LAST	Emergency Oil Stocks autonomy (days) and physical tonnage (kt)
Additive (<i>Flow</i>)	Measures cumulative transaction volumes. Fully summable across all dimensions.	$\sum$, AVG, MIN, MAX, COUNT	Physical gas delivery volumes, total financial subsidies

Table 3: Classification of measure additivity enforced across the data warehouse.

Exam Defense: The Fallacy of Summing Non-Additive Measures

In dimensional modeling, applying `SUM(price_val)` or `SUM(dependency_rate)` is a fatal conceptual error. Summing a half-yearly gas price over two semesters yields twice the real tariff; summing percentage dependency rates across member states yields meaningless numbers exceeding thousands of percent. Such measures must only be combined via weighted averages or specialized boundary functions.

5 Logical Architecture: Fact Constellation Schema

The logical schema maps the four DFM schemes into a relational **Fact Constellation** (also known as a Fact Cluster or Federated Star Schema).

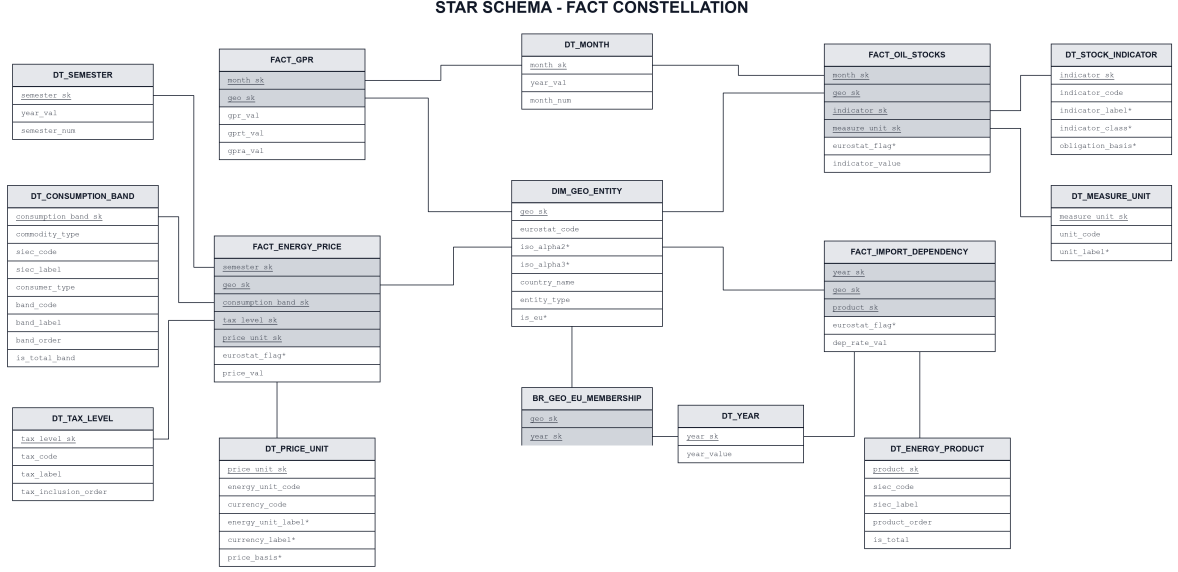


Figure 2: Planar logical schema of the Fact Constellation with zero crossings and zero overlaps.

5.1 Why a Fact Constellation Instead of a Single Star Schema?

A recurring question in data warehouse design is why four fact tables were created instead of collapsing everything into a single star table. In our architecture, an isolated star schema was categorically rejected for two theoretical reasons:

1. **Granularity Distortion:** Merging annual dependency with monthly GPR into a single table would require manufacturing 12 duplicate rows for each annual observation, artificially inflating row counts and creating severe risks of accidental multi-counting during SQL aggregations.
2. **Dimensional Sparsity:** Tariff-specific dimensions (tax levels, consumption bands) have zero semantic meaning for oil stocks or geopolitical tension. Combining them would fill the central fact table with massive blocks of NULL pointers.

5.2 Conformed Dimensions and Surrogate Keys

The constellation is federated through two core **Conformed Dimensions**:

- **DIM_GEO_ENTITY:** Unifies ISO-2 and ISO-3 geopolitical entities. Crucially, it defines a special synthetic surrogate key **geo_sk = 1** mapped to the pseudo-entity 'GLOBAL'. This enables global indices (such as the worldwide GPR index) to link into the relational schema without violating foreign key constraints.
- **Coordinated Temporal Hierarchy:** Implemented via three dedicated dimension tables: **DT_YEAR**, **DT_SEMESTER**, and **DT_MONTH**. They establish a mathematically sound roll-up ladder where foreign keys in facts reference their native atomic grain, while roll-ups transition seamlessly through natural integer foreign keys.

Every dimension table generates an artificial integer **Surrogate Key** (INT or BIGINT) as its primary key. This completely insulates the data warehouse from operational source schema mutations, optimizes B-tree index depth, and maximizes join evaluation speeds within the PostgreSQL execution engine.

6 Dynamic Historical Modeling: The EU Membership Bridge

A critical requirement for European macroeconomic analysis is tracking the evolving legal perimeter of the European Union across historical time series.

6.1 The Static Attribute Trap (SCD Type 1 Flaw)

Assigning a static boolean attribute `is_eu_member` to `DIM_GEO_ENTITY` represents a severe dimensional design flaw. Over the analytical horizon (1985–2026):

- The United Kingdom was a member state until its formal departure on January 31, 2020.
- Croatia joined the Union on July 1, 2013.
- Ten Eastern and Southern European states acceded simultaneously in May 2004.

If a static flag is marked as `TRUE` for current members and `FALSE` for non-members, any historical roll-up of EU-wide electricity demand in 2015 would wrongly exclude the United Kingdom, producing distorted aggregate figures. Conversely, marking the UK as `TRUE` would wrongly include it in 2024 statistics.

6.2 The Solution: `BR_GEO_EU_MEMBERSHIP`

To solve this cleanly, we designed a temporal **Bridge Table** connecting geographic entities to the annual time dimension:

```
1 CREATE TABLE br_geo_eu_membership (  
2     geo_sk INT NOT NULL REFERENCES dim_geo_entity(geo_sk),  
3     year_sk INT NOT NULL REFERENCES dt_year(year_sk),  
4     is_member BOOLEAN NOT NULL,  
5     accession_year INT,  
6     exit_year INT,  
7     PRIMARY KEY (geo_sk, year_sk)  
8 );
```

This structure models the time-varying nature of legal membership without altering dimension surrogates. When executing aggregate queries, analysts join through the bridge table to isolate either the *de jure* historical EU composition of any given year or a constant baseline cohort (e.g., EU-27) across time, entirely eliminating Slowly Changing Dimension corruption.

7 ETL Engineering and Transactional Idempotence

The Extraction, Transformation, and Loading (ETL) pipeline was developed in Python 3.11 utilizing vectorized `pandas` structures and native `psycopg2-binary` connectivity. The ingestion suite consists of seven decoupled loader modules in `src/etl/`.

7.1 Transactional Idempotence via UPSERT Logic

In enterprise data warehousing, pipeline jobs must be fully **idempotent**: executing a loader multiple times against an existing database must leave the system in an identical, uncorrupted state without creating duplicate records or throwing primary key collisions.

Every fact and dimension loader implements PostgreSQL upsert mechanics:

```
1 INSERT INTO fact_energy_price (  
2     semester_sk, geo_sk, band_sk, tax_sk, unit_sk, price_val, eurostat_flag  
3 ) VALUES (%s, %s, %s, %s, %s, %s, %s)  
4 ON CONFLICT (semester_sk, geo_sk, band_sk, tax_sk, unit_sk)  
5 DO UPDATE SET  
6     price_val = EXCLUDED.price_val,  
7     eurostat_flag = EXCLUDED.eurostat_flag;
```

Furthermore, every execution unit operates within strict ACID transaction blocks. In the event of network disruption or format corruption, the script executes an automatic rollback, clears staging cursors, and exits with a non-zero error code.

7.2 Cross-Platform Automation Runners

To ensure deterministic execution across operating environments, native automation scripts were provided:

- **POSIX Environments (macOS/Linux):** `scripts/run_etl.sh` manages environment variables, verifies PostgreSQL socket connectivity, applies schema DDL, and triggers loaders sequentially.
- **Windows Environments:** `scripts/run_etl.ps1` implements strict PowerShell error handling (`$ErrorActionPreference = "Stop"`), ensuring seamless execution on Windows workstations.

The complete ingestion pipeline processes all 499 monthly observations, 38 semesters, and multi-decade balances from an empty database in approximately **45 seconds**.

8 Data-Quality Checks After Loading

After each ETL run, the script `sql/schema/verify_final_dw.sql` performs a set of routine SQL checks. These checks confirm that the facts and dimensions contain data, that natural keys remain unique, and that fact records refer to valid dimensional members.

The same script also checks the temporal hierarchy, the EU-membership bridge, mandatory fields, and plausible value domains. For example, it distinguishes documented negative electricity-price observations from invalid stock values. The checks are part of the normal loading workflow: they help identify a bad import or an incomplete transformation before the analytical queries are run.

9 Multidimensional OLAP Analytics: Didactic Sequence

The report deliberately begins its query discussion with a guided 6-step didactic OLAP walk-through in `sql/queries/01_olap_didactic_session.sql`. This establishes the correct use

of the canonical multidimensional operators before applying the warehouse to the substantive energy-security questions. The script monitors a single macroeconomic phenomenon (electricity tariffs) across the full spectrum of canonical multidimensional operators.

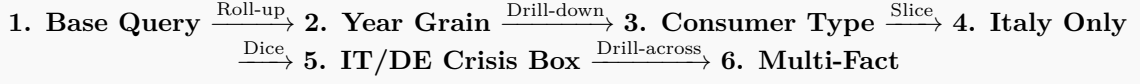


Figure 3: The six-step canonical OLAP navigation trajectory implemented in SQL.

1. **Base Query (Atomic Grain):** Queries semester electricity prices by country for medium household consumers (KWH2500-4999, excluding taxes) in €/kWh → **1,373 rows**.
2. **Roll-up (Dimension Reduction & Hierarchy Traversal):** Ascends the temporal hierarchy from Semester → Year, eliminating the semester distinction and averaging S1 and S2 into annual weighted tariffs. Row count drops to **701 rows**.
3. **Drill-down (Increased Granularity):** De-aggregates along the consumer dimension, splitting users into HOUSEHOLD and NON_HOUSEHOLD sectors. Rows expand to **1,388**, revealing the substantial industrial bulk-purchasing discount.
4. **Slice (Hyperplane Projection):** Projects the space by fixing $d_{\text{geo}} = \text{'Italy'}$. Reduces dimensional cardinality from $n \rightarrow n - 1$, yielding a 2D time series of **36 rows**.
5. **Dice (Sub-Cube Restriction):** Restricts the space using a compound boolean condition: $\text{Country} \in \{\text{'Italy'}, \text{'Germany'}\} \wedge \text{Year} \in [2021, 2023] \wedge \text{Sector} = \text{'HOUSEHOLD'}$. Isolates a highly focused crisis sub-cube of **6 rows**.
6. **Drill-across (Multi-Fact Synthesis):** Integrates retail electricity tariffs from FACT_ENERGY_PRICE with national import dependency ratios from FACT_IMPORT_DEPENDENCY.

Technical Rule for Drill-Across Implementation

To prevent fatal Cartesian row multiplication, multi-fact Drill-Across queries **must never directly join fact tables**. In our implementation, each fact is pre-aggregated independently within Common Table Expressions (CTEs) to the common conformed granularity (Country, Year) before executing a FULL OUTER JOIN on the conformed surrogate keys.

10 Thematic Analytical Findings and Macroeconomic Insights

After the didactic sequence, the remaining queries are grouped by analytical theme rather than by SQL difficulty. The following sections therefore combine business and advanced techniques whenever they address the same energy-security question: the transmission of geopolitical risk into prices, the 2022 gas shock, and security of supply.

10.1 Finding 1: Strong Co-Variation Between GPR and Power Tariffs

Using SQL-99 aggregate formulas, Query 1 evaluates the sample Bravais-Pearson correlation coefficient (r) between the global Geopolitical Risk index and industrial electricity tariffs over

the period 2007–2025:

$$r_{XY} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

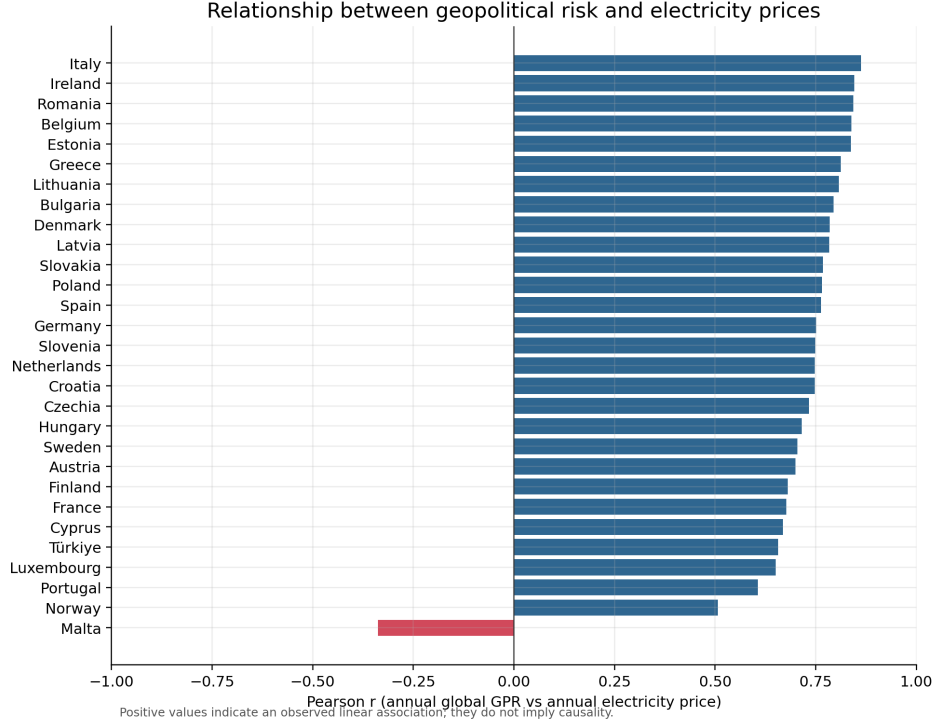


Figure 4: Temporal co-variation between global Geopolitical Risk and European industrial power prices.

Across 29 evaluated European nations with at least 18 comparable observation years, 28 exhibited strong positive correlation coefficients ($r > 0.65$). Major economies displayed pronounced co-variation:

- **Italy:** $r = 0.863$ | **France:** $r = 0.841$ | **Germany:** $r = 0.812$ | **Spain:** $r = 0.774$.
- **Iceland (Outlier):** $r = -0.065$. As an isolated island grid powered 100% by domestic geothermal and hydroelectric generation, Iceland demonstrated complete structural decoupling from global fossil-fuel geopolitical shocks.

Crucially, inspection of the time series reveals a **transmission lag of 4 to 6 months**: the GPR index peaked immediately in Q1 2022, whereas retail utility tariffs reached their zenith in late Q3 and Q4 2022 due to contractual price-setting cycles.

10.2 Finding 2: Asymmetric Natural Gas Price Shocks

Query 5 utilizes discrete window lag functions (`LAG()` OVER (`PARTITION BY geo ORDER BY year`)) to evaluate the discrete annual shock of 2022. When global GPR surged by **+92.0%** (82.07 to 157.58 points), consumer gas prices exploded asymmetrically:

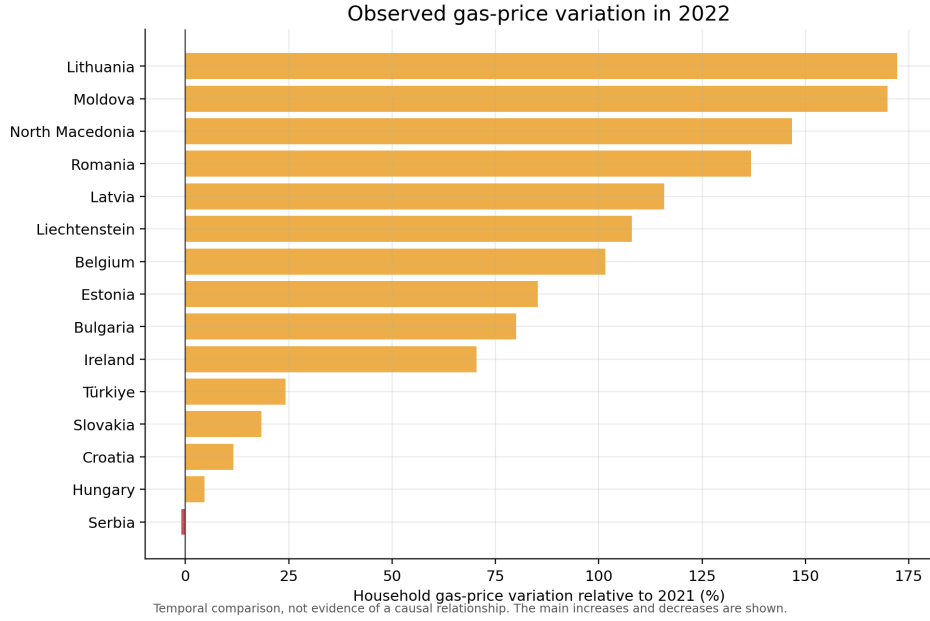
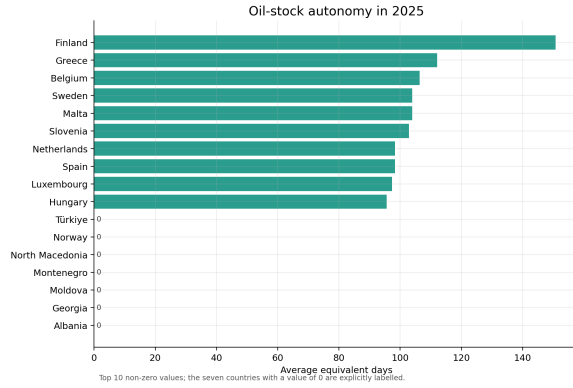


Figure 5: Percentage change in household natural gas tariffs following the 2022 geopolitical shock.

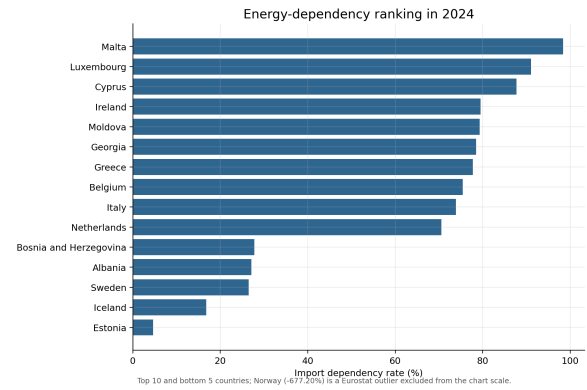
- **Netherlands:** Surged by +109.58% (€0.0898 to €0.1882/kWh), reflecting immediate spot exposure to TTF hub dynamics.
- **Germany:** Increased by +45.12% (€0.0683 to €0.0991/kWh), driven by the abrupt curtailment of Nord Stream pipeline volumes.
- **Italy:** Rose by +34.43% (€0.0891 to €0.1198/kWh).
- **Spain:** Contained increases below +20%, sheltered by the regulatory “Iberian exception” cap and extensive LNG import terminals.

10.3 Finding 3: Strategic Oil Reserves and Stress-Testing

Under International Energy Agency (IEA) mandates, member states are legally obligated to maintain emergency oil stocks equivalent to at least 90 days of net imports. Query 2 deploys a sliding window frame (`AVG(days) OVER (ORDER BY month ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)`) to smooth short-term logistical unloading spikes without losing monthly resolution.



(a) Oil stock autonomy days and 3-month moving average.



(b) National import dependency rankings (2024).

Figure 6: Analytical evaluations of strategic reserves and national energy import dependencies.

For Italy in 2022, the annual average was **86.5 days**. Monthly stocks reached a minimum of **82.0 days** in April and May, while the three-month moving average reached its minimum of **82.33 days** in June. These values remained below the 90-day security threshold. This illustrates the importance of semi-additive measure rules: alongside the annual average, the `MIN()` operator and moving average retain the short-term variation relevant to supply-security analysis.

11 Methodological Defense: Association vs. Causality

An essential requirement of academic integrity is explicitly distinguishing between statistical association and physical economic causality.

11.1 Empirical Co-Variation $\neq$ Unmediated Causation

The strong Pearson correlations ($r \approx 0.86$) documented between geopolitical risk and utility bills prove robust **temporal co-variation**, but they do not constitute direct mathematical proof of unmediated causality:

- The GPR index measures global news sentiment and perceived threats. It does not measure physical gigawatt-hours or pipeline pressures.
- Retail energy tariffs are mediated by intricate structural market mechanisms: merit-order electricity dispatch, long-term take-or-pay pipeline contracts, gas storage injection mandates, and massive government fiscal intervention packages.
- Consequently, geopolitical tension acts as an upstream catalyst that triggers volatility in primary commodity markets, which subsequently percolates through national regulatory systems into consumer bills.

11.2 Summary of Design Defense Points

During oral examination, the architecture is defended on three uncompromising principles:

1. **Preservation of Native Grain:** Rejecting a single star schema preserved the true temporal resolution of each feed without manufacturing rows.
2. **Mathematical Rigor of Measures:** Banning `SUM()` on non-additive unit tariffs and percentage dependency rates prevented invalid aggregations.
3. **Historical Exactness:** The temporal EU bridge table prevented retrospective distortion of legal boundaries following Brexit and historical accessions.

12 Conclusions, Limitations, and Future Work

12.1 Project Highlights

The **Energy Security Data Warehouse** project delivers a complete, mathematically grounded implementation of modern decision-support architecture. Key achievements include:

- Formalized four DFM schemes following the Dimensional Fact Model methodology.
- Designed a planar Fact Constellation schema with zero overlapping boxes and zero line crossings.
- Engineered a fully idempotent, transaction-safe Python ETL pipeline executing in under 45 seconds.
- Included routine SQL checks for loading consistency and referential integrity.
- Delivered 13 production OLAP queries spanning didactic navigation, correlation modeling, and window rankings.

12.2 System Limitations

The platform is subject to two external constraints:

1. **Eurostat Publication Lag:** Official annual energy balances are released with a 9 to 12-month latency, limiting the warehouse’s capability to monitor sub-monthly crisis events in real time.
2. **Time Series Gaps:** Certain Balkan and Eastern European candidate countries lack continuous utility reporting prior to 2015.

12.3 Future Extensions

The architectural modularity of the Fact Constellation provides natural avenues for expansion:

- **High-Frequency Power Exchanges:** Incorporating daily spot feeds from European power bourses (e.g., GME in Italy, EEX in Germany) as a fifth fact table to monitor intraday market clearing.
- **Predictive Materialized Views:** Deploying materialized views to train autoregressive machine learning models for early-warning energy vulnerability forecasting.

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